

Particulate Matter

Monitoring & Health Effects

Issue 2026-09-04 | Entry window 4 September 2026 (single day) | 27 records in scope, 55 screened out

27

RECORDS IN SCOPE
(55 SCREENED OUT)

9

SUBTOPIC CLUSTERS
OF TEN OCCUPIED

11

EFFECT ESTIMATES
(FIVE SOURCES)

6

SENSING /
INSTRUMENTATION

12

RECORDS WITH NO
HEALTH ENDPOINT

Signal of the day

Two papers, two countries, one result: the two quantities a PM instrument actually measures — scattered light and gravimetric mass — both decouple from the fraction that does the damage, and both fail in the direction that under-reports it.

Zeng et al. (*J. Hazard. Mater.*) put volume-normalised oxidative potential and light extinction on a common quadrant plane at two urban sites in southern Taiwan and isolate an “invisible killer” regime: **high redox activity, low optical impact**. The mediator is thermodynamic, not chemical — **high aerosol liquid water content** drives hygroscopic growth into optics-dominant states, while **low ALWC** corresponds to the highest OP-to-extinction ratios; secondary nitrate loads the extinction side specifically. Independently, Zhao C et al. (*Environ. Pollut.*) tracked 16 priority particle-bound PAHs across a real air-quality improvement in Kunming. Mass fell **32.7%** and BaP-equivalent burden **31.4%**, but the **particle-normalised toxicity, BaP_{eq}/PM_{2.5}, did not move** (-7.6% , $p=0.274$); 4–6-ring species stayed above **81%** of ΣPAH, and **40.3%** of simulated adult lifetime cancer risks still exceeded 10^{-6} . Put together: a nephelometric sensor reports the wrong thing when the air is dry, a gravimetric filter reports the wrong thing when the air is cleaner, and the error is the same sign in both cases. A network built to certify compliance against a mass threshold will read a genuine improvement while the per-microgram hazard is unchanged. Neither paper is a sensor study, and that is the point — this is the calibration target problem stated from the chemistry side.

What changed today

- ▶ **A normal-sized, unusually well-formed issue after two outliers.** 27 records against 38 on 3 September and 33 on 1 September. **24 distinct design labels over 27 records**, and for the first time in this run of issues **not one required a new entry in the design map** — 3 September needed fourteen. The corpus is broad without being ragged: *Measurement campaign* (7) leads, then *Review / synthesis* and *Observational — cohort* at 4 each.
- ▶ **The no-endpoint share fell hard, and that is a composition change rather than a trend.** 12 of 27 (44%) carry no human health endpoint, against 61%, 60% and 58% on the three preceding issues. The sensing block halved (11 → 6) while cohort epidemiology held, so the ratio moved because the *ES&T* by-ISSN deposit that inflated 3 September did not repeat.
- ▶ **Two UK Biobank cardiovascular papers, and their intervals are not on the same evidential footing.** Chen N et al. report per-IQR incident-VTE hazard ratios of **1.052 (1.005–1.101)** for elemental carbon and **1.090 (1.010–1.176)** for ammonium on 9,988 events — both lower bounds sit within 0.01 of the null, which is the honest width for a time-invariant residential exposure. The same paper’s heatwave hazard ratios run **1.149 (1.137–1.161)** to **1.407 (1.396–1.417)**: intervals roughly **three times narrower** on the same events, because a daily time-varying exposure carries far more information than an annual surface. Both belong in the paper; they should not be read off one forest panel as comparable quantities, and the mediation claim (elemental carbon and ammonium partly mediating the greenness effect) inherits the weaker exposure.
- ▶ **Two prenatal-PM cohorts with exactly inverted strengths.** Mbishi et al. have **119,110** Flemish children and municipality-scale kriged PM, and report a head-circumference deficit at 12 months of $-0.065 z$ **per IQR PM_{2.5}** — about one-fifteenth of a standard deviation, unambiguous at that n and, as the authors themselves write, small. Wei Q et al. have **2,680** pairs and weekly component-resolved exposure, and can name windows the first study cannot see: femur length most sensitive to black carbon at **gestational weeks 12–15**, head circumference to nitrate at **weeks 5–12 and 22–24**. Power and exposure resolution are being traded against each other, and no record today has both.
- ▶ **The widest source spread on the forest panel since August.** 11 estimates from **five** sources, tying 27 August; only the 26–27 July back-scan and 6 August have had more. One source still supplies 6 of the 11, so the panel is more diverse than it is balanced.
- ▶ **Geography narrowed.** China 8 of 27 (30%), *Global / multi-region* 7 (of which four are genuine multi-region reviews, one is a laboratory record and two are metadata-only records with no stated setting), East Asia ex-China 4. **Sub-Saharan Africa, South Asia, the Middle East**

and Oceania are all absent, three days after Sub-Saharan Africa tied its watch record at three. Latin America returns with a single Merida aerosol-chemistry record.

- **One new interventional trial registration, and it is a registry-hygiene warning.** NCT07805824 is a cluster RCT of a workplace heat and air-pollution mitigation package in **20 restaurants / 200 oven workers** in Najaf, Iraq, with **personal PM_{2.5} by gravimetric and real-time monitoring** as its sole secondary endpoint — directly relevant to occupational personal sensing. But it is filed as **Phase 4** with no drug, its start and primary-completion dates are both 2026-01-01 while its status is already COMPLETED, and the public record still carries the submitter's own pre-submission checklist asking them to "confirm the exact first-participant enrollment date". Treat the endpoint design as usable and the timeline fields as not.
- **Nil legs, one of them now verified rather than inspected.** Consensus is nil for the **sixth** consecutive run, but this time by DOI-level date check: its three low-cost-sensor calibration hits are all real and all out of window (Crossref created 2026-03-26 and 2026-04-21), and the third — the Alphasense/Sensirion recalibration-cadence paper — *is already in this corpus* from an earlier issue. The leg finds the right literature; only the date filter is missing. **OpenAlex** HTTP 429, unchanged since 28 July. **arXiv** answered cleanly for the second consecutive run, newest submission still 2026-08-20, contribution again zero.

Corpus analytics

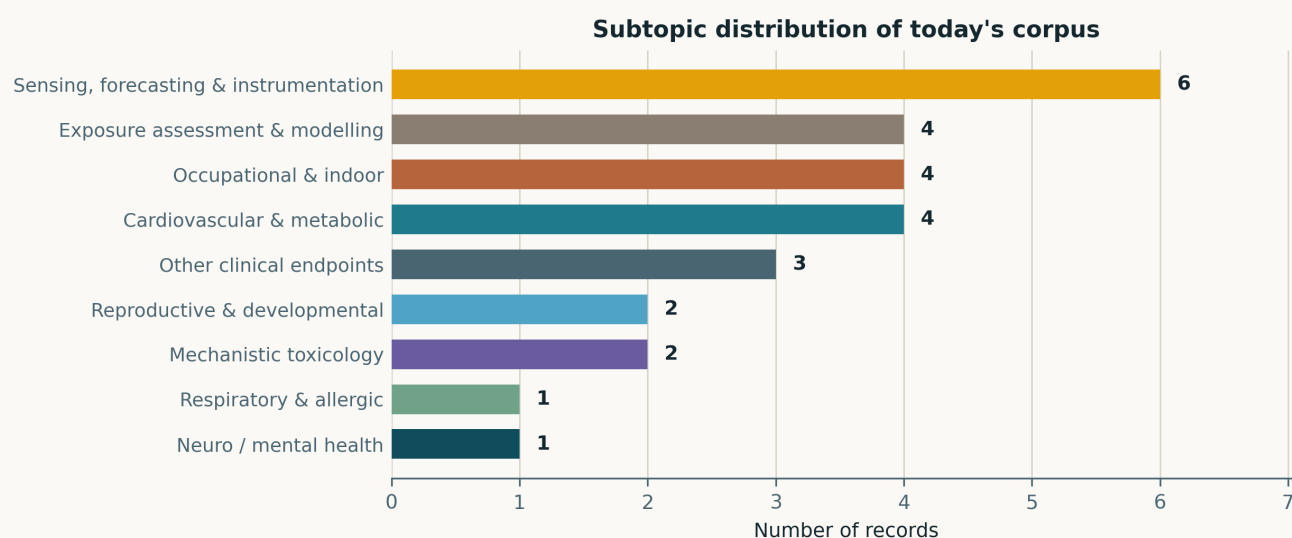


Fig. Subtopic assignment of the 27 in-scope records, single-label by dominant contribution. **Nine of ten clusters are occupied**, which has now happened on 12 of 41 issues; the vacant one is *Burden, policy & mitigation*, and it is vacant because today's two burden-adjacent records (US building-sector mortality, transboundary smoke exposure) both lead with an exposure surface and were filed under *Exposure assessment & modelling* instead. *Sensing* (6) and *Exposure* (4) hold 10 of 27, against 23 of 38 yesterday — the instrument-heavy run has broken.

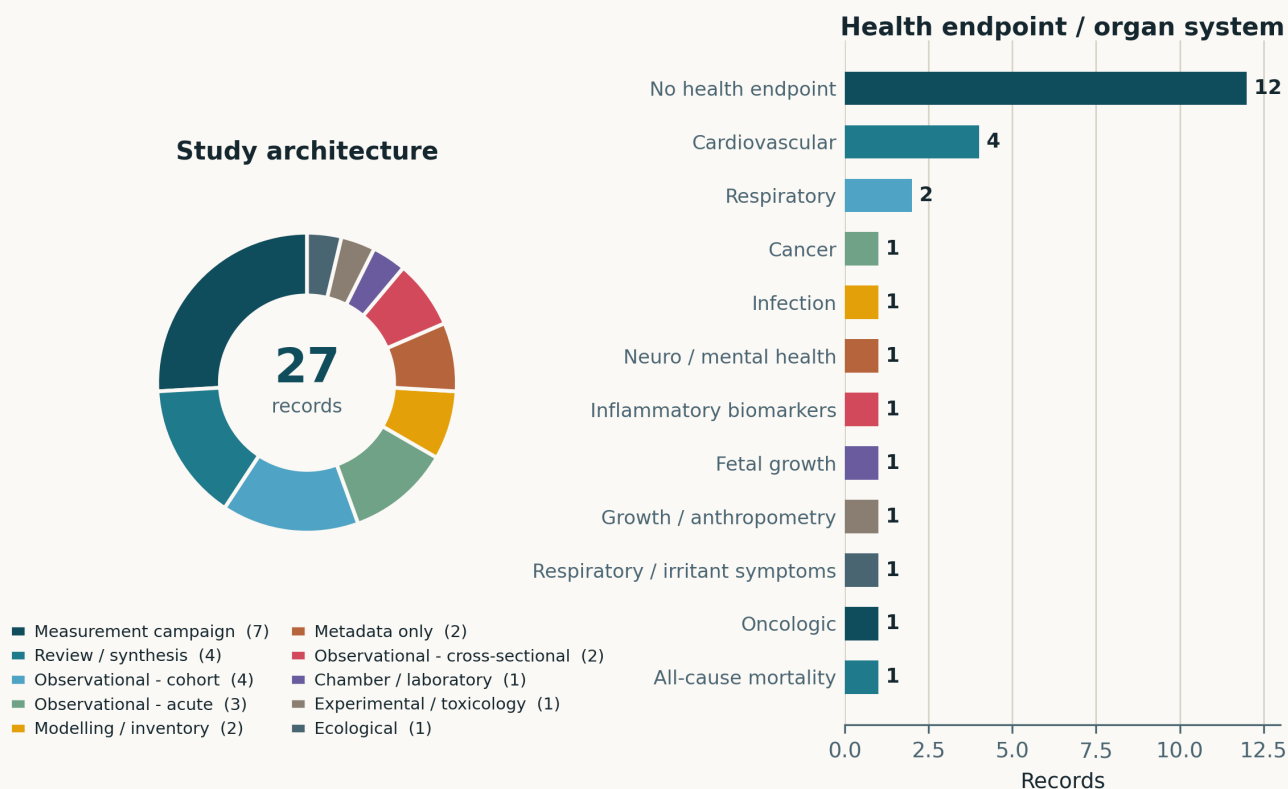


Fig. Left — study architecture over 24 distinct design labels, every one of which was already in `plots.py`'s design map; no map edit was needed this run, against fourteen additions on 3 September. *Measurement campaign* (7) leads, ahead of *Review / synthesis* and *Observational — cohort* at 4 each. Only 2 records ship as *Metadata only*, both Elsevier titles whose publishers deposited no abstract anywhere the pipeline can reach. Right — 12 of 27 (44%) carry no human health endpoint, the lowest share since 29 August and well below the 61% of the preceding issue. Behind that bar the health endpoints are unusually flat: none exceeds 4, and *Cardiovascular* reaches 4 only by pooling two cohorts, a time-series and a green-space cross-section. *Cancer* (1) and *Oncologic* (1) are one quantity drawn as two bars — an oesophageal incidence rate ratio and a computed carcinogenic-risk index — because `ENDPOINT_CANON` still does not merge them. That defect was logged on 3 September and is unfixed.

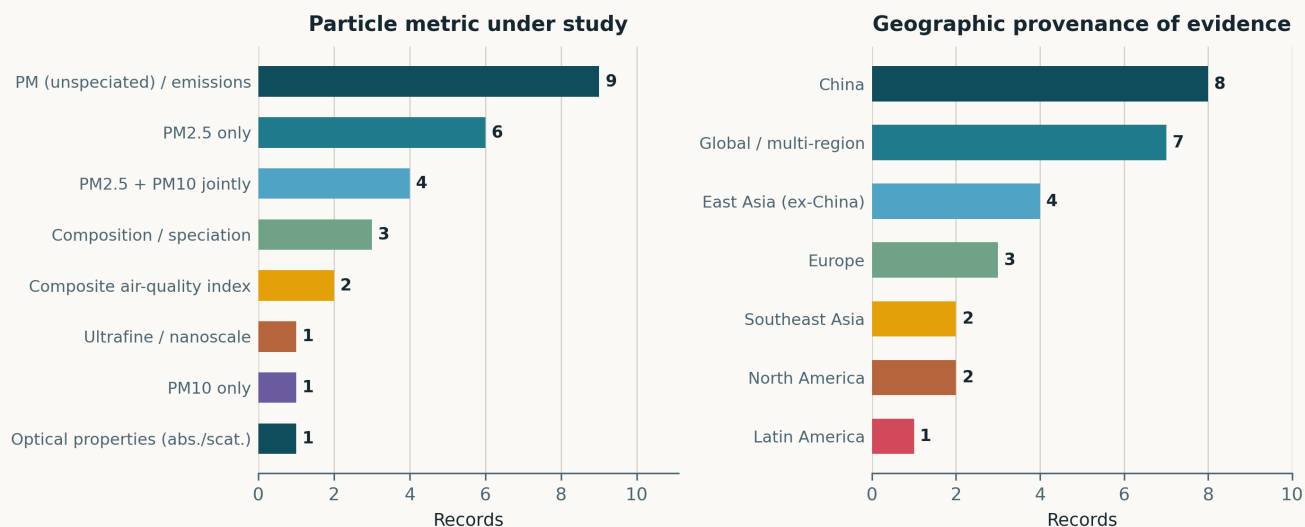


Fig. Left — particle metric. *PM (unspeciated) / emissions* at 9 carries the records whose measurand is not a regulated mass fraction: BVOC fluxes, odour activity values, bioaerosol counts, surface-attachment fractions and the two abstract-free records. *PM_{2.5} only* (6) plus *PM_{2.5} + PM₁₀ jointly* (4) is the epidemiological core; *Composition / speciation* (3) is where today's signal sits. Right — geography. China 8, *Global / multi-region* 7, East Asia ex-China 4 (two South Korean, one Japanese, one Taiwanese), Europe 3, North America 2, Southeast Asia 2, Latin America 1. **Sub-Saharan Africa, South Asia, the Middle East and Oceania are all empty.** North America at 2 continues a thin run — 1 on 27 August, 2 on 1 September, 8 on 3 September — so the 3 September figure was the anomaly, not this one.

Life-course windows addressed today (records may span several stages)

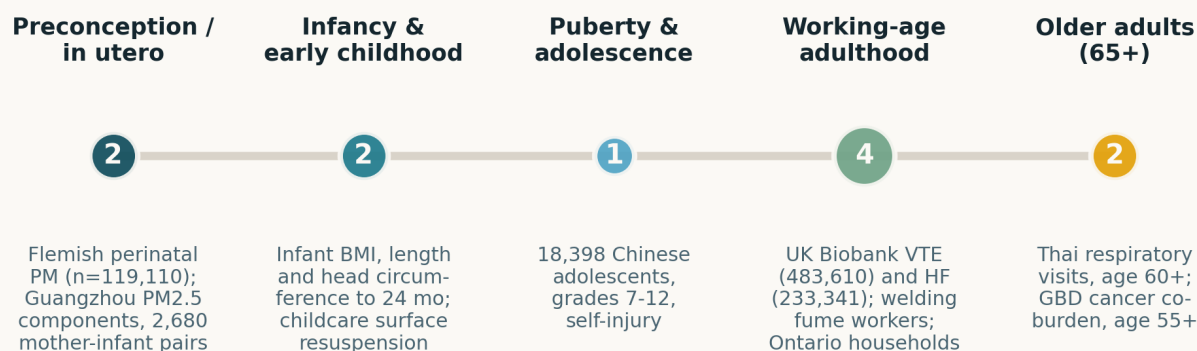


Fig. Life-course windows addressed; counts are non-exclusive and cover only records with a human endpoint or an exposed human population. Every stage is occupied, which is the first time since 27 August — puberty and adolescence, empty on the two preceding issues, is filled by a single record (18,398 Chinese adolescents). Working-age adulthood (4) leads, carried by the two UK Biobank cohorts, the welding-fume review and the Ontario household survey. “Older adults” is generous here: the two records are age ≥ 60 (Thailand) and age ≥ 55 (GBD), not 65+.

Where the work is happening: subtopic x study architecture

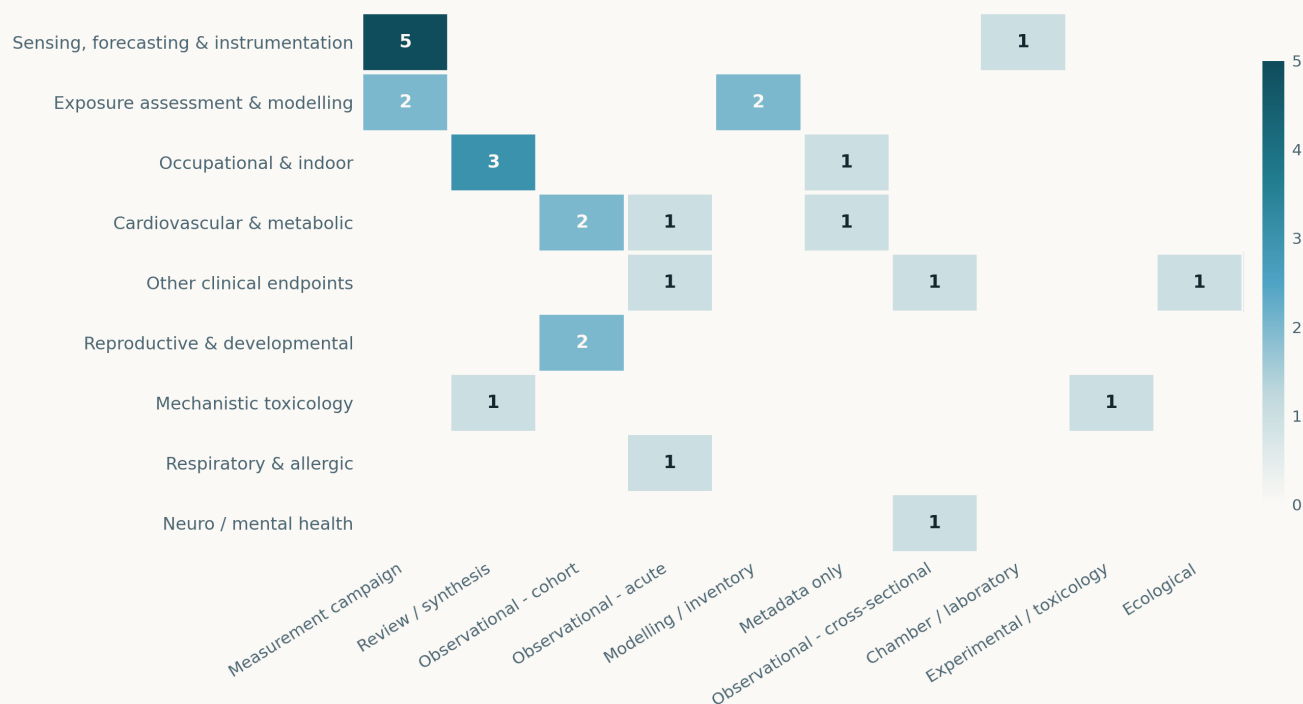


Fig. Subtopic $\times$ study-architecture cross-tabulation. The Sensing row is entirely Measurement campaign and Chamber / laboratory today — no modelling or forecasting records were filed under sensing, which is why the cluster is smaller but purer than yesterday's. Cardiovascular & metabolic and Other clinical endpoints are the only subtopics spread across three architecture families, and one of the three in the cardiovascular row is Metadata only, so part of that apparent breadth is an artefact of a missing abstract rather than real variety.

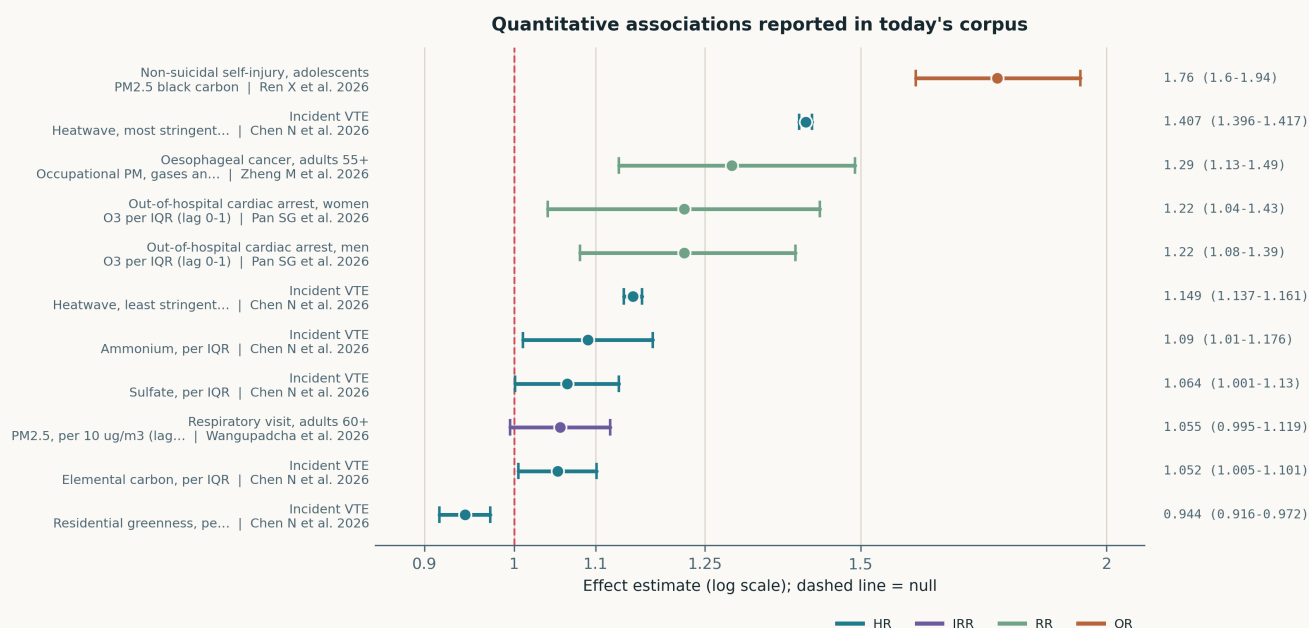


Fig. Eleven estimates from **five** sources — the widest source spread since 27 August. They are not commensurable and should not be read as a pooled panel. Six are from one UK Biobank VTE analysis and are internally correlated: three per-IQR $PM_{2.5}$ component hazard ratios whose lower bounds sit within 0.01 of the null, one protective greenness hazard ratio, and two heatwave hazard ratios whose intervals are roughly three times narrower because the exposure is daily rather than annual. The two ozone rate ratios for out-of-hospital cardiac arrest are identical point estimates in men and women (1.22) with different widths, which is what sex-stratification of 7,996 events buys and nothing more. The single IRR 1.055 (0.995–1.119) for respiratory visits crosses the null and is reported as null by its own authors. The black-carbon self-injury odds ratio 1.76 (1.60–1.94) is the largest effect on the panel and the weakest design — one cross-sectional survey round, exposure assigned by residence. Beta coefficients from the two fetal-growth cohorts are absent: a z -score mean difference has no place on a log-ratio axis.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Sensing, forecasting & instrumentation

6 records

Zeng et al. 2026d J Hazard Mater

Oxidative potential and light extinction decouple, and aerosol water is the switch

Paired oxidative-potential and optical measurements at two urban sites in southern Taiwan, arranged into a quadrant framework on *volume-normalised* OP against light extinction. Under coupling, mass-normalised OP and mass extinction efficiency share their drivers, principally **water-soluble organic carbon**. Under decoupling, **high aerosol liquid water content** pushes the aerosol into an optics-dominant state through hygroscopic growth, while **low ALWC** yields the highest OP-to-extinction ratios — the “invisible killer” quadrant. Secondary nitrate loads extinction specifically; aerosol pH is a weaker secondary term. The direct consequence for distributed sensing: every nephelometric low-cost sensor estimates mass from scattering, so its error against the redox-active fraction is not random but conditioned on humidity, in the same direction as its already-known RH bias. Weakness: OP is one assay on one aerosol population at two sites, sampling duration is not stated in the abstract, and no causal chain from OP to a clinical endpoint is tested here.

doi: 10.1016/j.jhazmat.2026.143459

Ichikawa et al. 2026 Atmos Chem Phys

Drone sampling agrees with a flux tower to within 30%, and that is the useful number

Multi-year warm-season BVOC measurements at multiple heights on a 30 m flux tower in a *Quercus serrata*-dominated suburban Tokyo forest, with supplemental drone sampling and comparison against MEGAN. Isoprene is **over 90% of measured BVOC composition** at peak summer, monoterpenes stay low with weak vertical gradients, and daily average isoprene emission fluxes span -0.05 to $15.30 \text{ mg m}^{-2} \text{ h}^{-1}$. The methodological result is the one worth carrying: **drone-measured horizontal variability is 10–30% within 30 m of the tower, and tower- and drone-derived flux estimates differ by about 30%**. That is a direct, published estimate of the representativeness error a mobile aerial platform introduces relative to a fixed mast — the same quantity a mobile PM platform needs and rarely reports. Limits: BVOCs are SOA precursors, not particles; intermittent campaigns over multiple years confound season with instrument configuration.

doi: 10.5194/acp-26-12521-2026

Kim C-h et al. 2026 Atmosphere

Block-missing monitoring data, five imputation models, and an honest sensitivity test

Eleven months of 20-minute measurements of 22 designated odour compounds at a livestock farm, a wastewater plant and an organic-waste plant in Eumseong, South Korea, with **multi-week to multi-month block gaps** — the failure mode that actually afflicts long-running low-cost networks, as opposed to the scattered single-point dropout most imputation papers assume. Compound-specific performance was assessed by **block holdout** against BiLSTM, BiGRU, TCN, Transformer and HistGradientBoosting. Odour activity values were computed *primarily from observed concentrations*, with the fully imputed series used only as a sensitivity comparison to quantify interpolation-induced bias, which is the correct discipline and is rarely followed. Ammonia was excluded outright for a low valid-observation rate. Odour contribution was dominated by volatile fatty acids and trimethylamine rather than hydrogen sulfide. Transferable to PM networks in method, not in analyte.

doi: 10.3390/atmos17090868

Cai J et al. 2026 Environ Sci Technol

Subway air microbiome is homogeneous at network scale and heterogeneous at platform scale

Stratified cluster sampling of **43 underground stations across 13 lines** of the Shanghai subway, collecting total suspended particles and PM_{2.5} *simultaneously* at entrances, halls and platforms, then integrating taxonomy, transcriptional potential, PM_{2.5} chemistry, microclimate and network topology. Bacteria dominate both size fractions and most transcriptional signal; eukaryotes are strongly enriched in TSP, **a clean demonstration of particle-size filtering of the biological load**. Dominant taxa (*Staphylococcus*, *Bacillus*) show weak abundance–transcription coupling, so abundance is a poor proxy for activity. Diversity and composition are essentially uniform across the network — consistent with passenger mixing — while finer-scale structure appears within stations. For exposure work this means a single station is a reasonable network sentinel for community composition and a poor one for microenvironment. Weakness: one city, one season not stated, no human outcome.

doi: 10.1021/acs.est.6c00174

Rosas et al. 2026 Environ Sci Pollut Res Int

The PM_{2.5}/PM₁₀ ratio separates two aerosol regimes that mass alone cannot

Continuous PM_{2.5}, PM₁₀ and particle-bound PAH at a near-coastal urban site in Merida, Mexico, joined to FIRMS fire detections and HYSPLIT back-trajectories, contrasting the March–May regional biomass-burning season against the June–August African dust season. Concentrations are the wrong discriminator: **PM_{2.5} and PM₁₀ rise only modestly during dust**, but the **PM_{2.5}/PM₁₀ ratio falls**, marking the coarse mineral contribution, while biomass burning loads the fine mode and the combustion tracers. A network reporting PM_{2.5} mass alone would score the dust season as the milder of the two and miss the toxicologically distinct composition entirely. Part I of a two-part study, so the health framing in the title is anticipatory rather than delivered here. Single site, single urban domain.

doi: 10.1007/s11356-026-38193-x

Seong & Hoque 2026 Front Pediatr

Attachment fraction ranks metal > wood > glass, and carpet is two microenvironments

Batch attachment and centrifugal detachment of *Corynebacterium* sp. on carpet, wood, metal and glass, with near-surface velocity fields resolved by particle image velocimetry and surfaces characterised for roughness, contact angle and porosity. Attachment fraction rose with exposure time on all materials, reaching **~0.59 on metal, ~0.48 on wood, ~0.36 on glass**. The useful observation is morphological: **carpet generates two distinct microenvironments** — a higher-velocity outer-fibre region and a sheltered low-velocity inter-fibre region — so a single resuspension coefficient for “carpet” is a category error. Wood produced the most spatially heterogeneous velocity field, glass the most uniform. Relevance to PM is indirect but real: resuspension source terms in indoor exposure models are usually fitted per material with no flow-field justification. One organism, no PM measurement, replicate count not stated.

doi: 10.3389/fped.2026.1895305

Exposure assessment & modelling

4 records

Ullah et al. 2026 Environ Pollut*Transboundary smoke moves almost nobody across China's standard and almost everybody across WHO's*

Daily PM_{2.5} from **51 cities** in Yunnan, Guangxi and Guangdong for March–April 2015–2019, gridded onto WorldPop, with the Indo-China Peninsula fire contribution isolated by paired GEOS-Chem runs with and without the fires. The population-weighted share of days above the **WHO 2021 24 h guideline (15 µg/m³) reached 78–96% every year**; above **China's Grade II standard (75 µg/m³) it was 1–6%**, an average daily exposed population of about **6 million**, under 3% of the corridor's residents. The two benchmarks describe different worlds from identical data, which is the paper's real contribution to monitoring policy. Principal weakness: the attribution runs at 2° × 2.5°, roughly 200 km, so the fire contribution is a regional average imposed on city-scale observations and cannot support any within-province ranking.

doi: 10.1016/j.envpol.2026.129081

Arunachalam et al. 2026 Research Square (preprint)*Building-sector combustion mortality is three times prior estimates, and it is mostly NO₂*

CMAQ Decoupled Direct Method sensitivities of NO₂, O₃ and PM_{2.5} to commercial and residential building-sector precursor emissions for the contiguous US (2021), propagated through BenMAPR. Net burden **≈14,000 premature deaths per year (95% CI 7,900–19,000): 12,000 from NO₂, 3,100 from PM_{2.5}, and 1,000 deaths avoided** through O₃ reductions. Residential gas alone accounts for **≈7,900**. The headline “three times higher than previously reported” is carried almost entirely by the NO₂ term, so the result is a statement about which NO₂ concentration–response function you accept, not about building emissions being larger than thought. Preprint, not peer reviewed. The negative O₃ term is the sign of a titration effect and is correctly reported rather than suppressed, which is a point in the paper's favour.

doi: 10.21203/rs.3.rs-10758198/v1

Zhao C et al. 2026 Environ Pollut*Mass fell a third; toxicity per microgram did not move*

Sixteen priority PM_{2.5}-bound PAHs in Kunming, compared between a higher- and a lower-PM_{2.5} stage. Raw declines: **PM_{2.5} –32.7%, ΣPAHs –27.4%, BaP_{eq} –31.4%**; covariate-adjusted, **–28.8% and –33.5%**. But the particle-normalised intensity **BaP_{eq}/PM_{2.5} changed by –7.6%** ($p=0.274$) and **BaP_{eq}/ΣPAH** by only **–6.5%** ($p=0.026$). Four-to-six-ring species held above **81%** of ΣPAH and BaP plus dibenzo[a,h]anthracene supplied over **82%** of BaP_{eq} in both stages; **40.3%** of simulated adult incremental lifetime cancer risks still exceeded 10^{–6}. PMF patterns are read conservatively as source-related rather than as resolved sources, which is the right caution. The inference is threatened by the two-stage design: “stage” bundles season, meteorology and policy, and no counterfactual separates them.

doi: 10.1016/j.envpol.2026.129094

Leung et al. 2026 Indoor Air*Abandoned wells, indoor VOCs, and a pilot honest enough to report its own limits*

Passive indoor sampling for 13 VOCs in **17 households (October 2024)** and **15 households (February–March 2025)** in southwestern Ontario, with abandoned-well exposure estimated by inverse-distance-squared weighting over wells within 3 km and self-reported acute symptoms over the previous six months and during sampling. This is gas-phase, not particulate — carried here because the exposure-assignment design is the one a distributed indoor PM study would use and because the MAPSH project measures both. At $n<20$ per round with Spearman correlations and no personal monitoring, the study is powered to describe indoor concentrations against Canadian guidelines and not to detect a well-proximity association; the six-month symptom recall window compounds that. Read as a feasibility and concentration-range report.

doi: 10.1155/ina/8041546

Occupational & indoor**4 records****Ibrahim 2026b** Qeios (preprint)*The case for routine indoor PM monitoring, argued without data*

A commentary arguing that Malaysia's ICOP IAQ 2010 framework, though it contains preventive elements, is operated as a complaint-triggered investigation regime, and that **particulate matter and outdoor-to-indoor pollutant transport** should join CO₂ as routinely monitored parameters in building management. The specific climatic argument — that hot, humid conditions couple ventilation, condensation and biological growth in ways a temperate-derived code handles badly — is sound and underexploited in the LCS literature. Everything else is assertion: single author, non-peer-reviewed platform, no primary data, and the Malaysian office, hospital and school studies it cites are described rather than synthesised. Included as an argument about monitoring design, not as evidence.

doi: 10.32388/b506md

Wei J et al. 2026 Front Immunol*Welding fume immunology, with the exposure-assessment gap named explicitly*

Mini review of metal-rich fine and ultrafine welding-fume particles as a stimulus to airway epithelium, alveolar macrophages, neutrophils, monocytes and circulating immune cells. Consistent signals across human exposure studies, controlled inhalation, cell models and workplace investigations: acute-phase proteins, cytokines, chemokines and neutrophilic responses, with some evidence of altered T-helper activity and blunted responsiveness to bacterial stimuli. TRPA1/TRPV1 sensory signalling is proposed as a rapid route to cough and airway reflex, and the review is careful to say that direct welding-fume-specific human evidence for that pathway is limited. The stated obstacle to synthesis is the one that matters here: **fume composition and particle metric vary so much between studies that biomarker results cannot be pooled**. No quantitative estimate, no pooling, no systematic search protocol reported.

doi: 10.3389/fimmu.2026.1882845

Kusi et al. 2026 Research Square (preprint)*Waste-tyre combustion toxicology: a systematic search that could not pool*

PRISMA-style search of PubMed, Scopus, ScienceDirect and Google Scholar for 2016–2025 human and experimental studies of waste-tyre combustion emissions — particulate matter, PAHs, VOCs and heavy metals — reporting biochemical, molecular, biomonitoring or toxicological outcomes, appraised with SYRCLE, JBI and NIH instruments as appropriate to design. **Heterogeneity in exposure conditions, designs and reported outcomes precluded quantitative synthesis**, which the authors state plainly. The value is in mapping organ-specific mechanistic pathways for an exposure that is ubiquitous in informal waste economies and almost absent from regulated monitoring networks. Preprint, not peer reviewed; a narrative synthesis of a heterogeneous literature inherits every selection effect in that literature.

doi: 10.21203/rs.3.rs-10530000/v1

Derwein et al. 2026 Build Environ*Masks, filtration and ventilation on exhaled-air transmission — metadata only*

No abstract was deposited to Crossref, Europe PMC or Semantic Scholar at index time, and ScienceDirect refused retrieval from this environment, so nothing beyond title, authorship and venue can be asserted. Recorded so the DOI enters `seen.json` and the record is not re-screened; queued for the weekly abstract re-check. Listed under *Occupational & indoor* on the title's face, which is a provisional assignment.

doi: 10.1016/j.buildenv.2026.115206

Cardiovascular & metabolic

4 records

Chen N et al. 2026 Eur J Prev Cardiol

Elemental carbon leads the PM_{2.5} components on venous thromboembolism, barely

483,610 UK Biobank participants (mean age 56.5, 54.3% female), median 11.83 years, 9,988 incident VTE from hospital and death records. Per-IQR hazard ratios: elemental carbon 1.052 (1.005–1.101), sulfate 1.064 (1.001–1.130), ammonium 1.090 (1.010–1.176); organic matter and nitrate not significant. Residential greenness 0.944 (0.916–0.972), partly mediated by elemental carbon and ammonium. Heatwave hazard ratios span 1.149 (1.137–1.161) to 1.407 (1.396–1.417) across definitions. Two cautions. The three positive component intervals all have lower bounds within 0.01 of unity at 3×3 km resolution, so component ordering is not established. And the heatwave intervals are roughly three times narrower on the same events because that exposure is daily rather than annual — a real difference in information, not a real difference in certainty about air pollution.

doi: 10.1093/eurjpc/zwag465

Fang et al. 2026 Eur J Prev Cardiol

Lifestyle matters more where the air is worse, and most for early-onset heart failure

233,341 UK Biobank participants, 2006–2010 baseline to 2023, with an air pollution score over NO₂, NO_x, PM₁₀, PM_{2.5} and PM_{2.5–10} crossed against a six-component lifestyle score. Additive interaction is present and larger for early onset: RERI 0.44 (0.14–0.74) for all heart failure, 1.33 (0.03–2.62) early-onset, 0.39 (0.08–0.70) late-onset, with the early-versus-late contrast at $p=0.08$ — i.e. the headline age difference is not itself statistically resolved, and the early-onset interval very nearly touches zero. Read the other direction instead, where the contrast is firmer: lifestyle differences map to 3.35-fold (1.76–9.79) larger early-onset incidence differences under high pollution against 1.52-fold (1.33–1.74) under low. A composite pollution score cannot attribute anything to particles specifically.

doi: 10.1093/eurjpc/zwag456

Pan SG et al. 2026 Front Public Health

Ozone, not PM, survives two-pollutant adjustment against cardiac arrest — in one city-year

7,996 out-of-hospital cardiac arrests from the Tianjin prehospital emergency system, July 2023–June 2024 (mean 21.9/day), with a quasi-Poisson DLNM over lag 0–14 for temperature and lag0–1 single- and two-pollutant models for PM_{2.5}, PM₁₀, NO₂, SO₂, CO and O₃. **Ozone was the only pollutant consistently associated:** per-IQR lag0–1, +15.3% overall, men RR 1.22 (1.08–1.39), women 1.22 (1.04–1.43). The temperature result is reported with unusual honesty — a monotonic cumulative risk curve, minimum-mortality temperature –1.9 °C, apparent heat peak at lag 1 (RR ≈1.57 at the 95th percentile) whose confidence intervals include unity, and the authors say so in the abstract. The binding limit is exposure length: **one city, twelve months**, so season and pollutant are almost fully confounded and the null PM result is uninformative rather than negative.

doi: 10.3389/fpubh.2026.1871041

Jiang Z et al. 2026 Atmos Environ

Multipollutant exposure and cardio-renal-metabolic multimorbidity trajectories — metadata only

No abstract was deposited to Crossref, Europe PMC or Semantic Scholar at index time. Design, cohort, exposure model, sample size and setting are all unknown and are recorded as such rather than inferred from the title. It is the 32nd *Atmospheric Environment* record to enter this corpus abstract-free since 1 August, which makes this a chronic publisher deposit gap rather than an incident; the DOI is queued for the weekly re-check, and the record occupies a *Cardiovascular & metabolic* slot on the title's face only.

doi: 10.1016/j.atmosenv.2026.122340

Reproductive & developmental

2 records

Mbishi et al. 2026 Environ Pollut

Perinatal PM and infant head circumference: unambiguous, and small

119,110 Flemish children with BMI-, length- and head-circumference-for-age z -scores from routine well-baby clinic records, perinatal PM₁₀ and PM_{2.5} from the RIO detrended-kriging model, and mixed-effects models carrying a 5-df natural cubic spline in child age with exposure-by-age interaction and child-level random intercepts and slopes. At 12 months, per IQR: head circumference –0.0499 (95% CI –0.0606, –0.0393) z for PM₁₀ and –0.0645 (–0.0771, –0.0518) for PM_{2.5}, with inverse associations sustained between 6 and 24 months. BMI- and length-for-age associations change sign across age, and green-space associations are inconsistent across measures. The authors state that estimates are small and that most nominally significant associations are of limited magnitude — roughly a fifteenth of a standard deviation. Exposure is municipality-scale interpolation, which at this n makes precision a poor guide to accuracy.

doi: 10.1016/j.envpol.2026.129064

Wei Q et al. 2026 Int J Epidemiol

Weekly component windows: black carbon at 12–15 weeks, nitrate at 5–12 and 22–24

2,680 mother–newborn pairs and 9,339 growth ultrasounds (abdominal circumference, head circumference, femur length, estimated fetal weight), Guangzhou 2017–2020, with weekly black carbon, organic matter, nitrate, sulfate and ammonium from the TAP model entered into distributed lag models with GEE, plus machine-learning mixture analysis. Femur-length reduction is most strongly associated with **black carbon at gestational weeks 12–15**; head-circumference deficits with **nitrate at weeks 5–12 and 22–24**; estimated fetal weight with **nitrate at weeks 6–8**. Black carbon and nitrate dominate the mixture. This is the exposure resolution the Flemish cohort above lacks, at 45× fewer children. The threat to inference is that five components drawn from one gridded product co-vary strongly, so “which component” is partly a question about the exposure model rather than about biology; the abstract reports no effect sizes for the identified windows.

doi: 10.1093/ije/dyag185

Mechanistic toxicology

2 records

Heo et al. 2026 Tuberc Respir Dis (Seoul)

Three weeks of PM₁₀ at the air–liquid interface: barrier down, differentiation lost, Notch up

Differentiated human small airway epithelial ALI cultures exposed to PM₁₀ for 7, 14 and 21 days. Transepithelial electrical resistance **fell progressively**; club-cell and ciliated-cell markers fell while basal-cell markers were preserved, i.e. a differentiation block rather than uniform cytotoxicity. Surfactant **SP-A and SP-B mRNA fell**, yet **secreted SP-A rose while SP-B secretion fell** — a dissociation the paper reports without over-explaining. IL-1 β , IL-6, IL-8 and TNF- α rose; cleaved Notch1 and Notch3, phospho-NF- κ B, I κ B α , HES1 and HEY1 all increased. The chronic, repeated-dose ALI design is a genuine advance on single-dose submerged culture. Limits are the usual ones and are not minor: **PM₁₀ dose, source and donor number are not stated in the abstract**, and no pathway inhibition experiment establishes that Notch is causal rather than concurrent.

doi: 10.4046/trd.2026.0072

Ren Y et al. 2026 Research (Wash D C)

Lung-on-chip for PM, appraised model by model

Critical review of *in vitro* platforms for PM–lung interaction, from 2D monolayers through 3D bioprinted constructs and organoids to lung-on-chip devices, with each class assessed for what it can and cannot reproduce. The central argument is that breath-mimicking cyclic mechanical strain is not an incidental refinement but a determinant of how particles interact with the epithelial interface — deposition, clearance and mechanotransduction all change under motion, so static models systematically misrepresent the dose actually delivered. Useful as an orientation to which platform answers which question. It is a narrative review with no systematic search, no pooled comparison and a closing pitch for the authors' own “highly integrated and intelligent” platform, so the appraisal is not disinterested.

doi: 10.34133/research.1414

Neuro / mental health

1 record

Ren X et al. 2026 Front Public Health

Black carbon dominates a PM_{2.5} mixture on adolescent self-injury — in cross-section

18,398 adolescents in grades 7–12 across 15 schools in four Chinese provinces, with non-suicidal self-injury by ANSAQ and multidimensional biological rhythm by SBRDA. PM_{2.5} and its constituents were positively associated with NSSI risk, black carbon most strongly: **OR 1.76 (1.60–1.94)**, and dominant in the mixture models (**WQS weight 0.810**, Qgcomp effect 0.671). This is the largest effect on today's forest panel and rests on the weakest design there. A single cross-sectional survey round cannot order exposure and outcome in time; exposure is assigned by school or residence, so the black-carbon signal is also a traffic-proximity and urbanicity signal; and rhythm disorder is plausibly a mediator, a confounder or a consequence, which stratification cannot distinguish. The mixture weight is the informative part — it says the components are not interchangeable — and the odds ratio is not the part to quote.

doi: 10.3389/fpubh.2026.1887542

Respiratory & allergic

1 record

Wangupadcha et al. 2026 J Glob Health

A null result, reported as null, at 42 μ g/m³

Time-series of **924** respiratory visits (ICD-10 J00–J99, mean 6.37/day) among residents aged ≥ 60 living within 5 km of a newly established industrial area in central Thailand, November 2023–March 2024, against daily PM_{2.5} from the nearest government monitor, with negative binomial single-lag and moving-average models adjusted for temperature, humidity, wind speed, day of week and trend. Mean PM_{2.5} was **42.43 μ g/m³ (SD 15.12)**, far above WHO guidance. The largest estimate was at lag 4: **IRR 1.055 (0.995–1.119)** per 10 μ g/m³ — positive, not significant, and described that way by the authors rather than spun. Subgroup estimates were numerically larger in women and at 60–69 with non-significant interaction tests. The study is underpowered by construction: **five months, 924 events, one monitor** for a 5 km radius. Worth carrying as a well-behaved null and as a monitoring-density argument.

doi: 10.7189/jogh.16.04269

Other clinical endpoints

3 records

Ding E et al. 2026 Environ Sci Technol

Compound climate extremes and influenza, with a submultiplicative interaction

630,120 laboratory-confirmed influenza cases from hospitals in **335 Chinese cities**, 2005–2019, analysed by case-crossover with distributed lag non-linear models, machine-learning variable ranking, and an LSTM forecasting proof of concept for Beijing. Temperature, humidity and particulate matter rank as the primary contributors. The result worth keeping: co-exposure to multiple extreme events carries higher joint risk than single exposure, but the interaction is **submultiplicative on the multiplicative scale with no evidence of additive interaction** — so the joint effect is less than the product and the naive stacking of extreme-event risks overstates it. Adding extreme-event terms improved daily forecasting in the Beijing model. Caveats: the abstract reports no effect sizes at all, hospital-based case ascertainment varies across 335 cities and 15 years, and one-city forecasting is a demonstration, not a validation.

doi: 10.1021/acs.est.6c04755

Zheng M et al. 2026 Ann Med

Occupational particulate exposure ranks with alcohol in an ecological cancer model

GBD 2021 incidence for adults ≥ 55 across **203 countries**, 1990–2021, classified into quartile-based co-burden categories for oesophageal and lip/oral-cavity cancer, with 58 candidate exposures screened by random forest and SHAP to 36, then to 25 after variance-inflation screening, entering negative binomial models with population offsets. Countries split into consistent (73), oesophageal-dominant (61) and LOC-dominant (69). **Occupational exposure to particulate matter, gases and fumes carries RR 1.29 (1.13–1.49)** for oesophageal cancer, comparable to high alcohol use (1.32, 1.13–1.53) and above low nut/seed intake (1.16). The authors state the correct caveat themselves: these are **country-level ecological associations** and establish no individual-level causality. GBD exposure estimates for occupational PM are modelled, not measured, so the estimate is a model-on-model contrast.

doi: 10.1080/07853890.2026.2721134

Wang C et al. 2026 Environ Sci Technol

Green space and cardiovascular health where the air-quality pathway is 75% of the model and the effects are fragile

Two-stage gradient-boosting plus structural equation modelling on a clinical cohort of **23,312** patients in Nanjing. **Air-quality-related pathways account for more than 75% of the relative statistical contribution to model variance**, exceeding psychological-restoration proxies — which, if it holds, says green-space cardiovascular benefit in a dense polluted city is mostly an air-quality mechanism. Distinct networks emerge for chronic ischaemic heart disease (vegetation vitality inverse to NO₂ and SO₂, linked to metabolic markers) and heart failure. The paper is unusually candid about its own weaknesses and they are disqualifying for the headline: the macroscale aerodynamic interpretation is called speculative for want of morphological or fluid-dynamics data, and the age-stratified canopy-density × PM_{2.5} slopes are **nonsignificant with extremely small effect sizes**. Relative variance contribution is not effect size.

doi: 10.1021/acs.est.6c08296

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Ren X et al.	Association between PM _{2.5} constituents and non-suicidal self-injury among Chinese adolescents: modifying role of biological rhythm	Front Public Health	Cross-sectional survey	PM _{2.5} total mass and constituents in logistic and weighted quantile sum regression, stratified by multidimensional biological-rhythm score	China
Cardiovascular & metabolic					
Chen N et al.	Associations of PM _{2.5} components, residential greenness, and heatwaves with incident venous thromboembolism	Eur J Prev Cardiol	Prospective cohort (mixtures)	Elemental carbon, organic matter, sulfate, ammonium and nitrate at 3x3 km; NDVI in a 300 m buffer; heatwaves under several temperature thresholds	United Kingdom
Fang et al.	Joint association of air pollutants and lifestyles on early-/late-onset heart failure: a population-based prospective cohort study	Eur J Prev Cardiol	Prospective cohort	Composite air-pollution score over NO ₂ , NO _x , PM ₁₀ , PM _{2.5} and PM _{2.5-10} , crossed with a six-component lifestyle score; multiplicative and additive interaction	United Kingdom
Jiang Z et al.	Long-term exposure to multiple air pollution and trajectory risk of cardiovascular-metabolic multimorbidity	Atmos Environ	Metadata only (no abstract)	Not retrievable: the publisher deposited no abstract to Crossref, Europe PMC or Semantic Scholar at index time	Not stated (metadata only)
Pan SG et al.	Ozone and possible heat effects on out-of-hospital cardiac arrest: a time-series study	Front Public Health	Distributed-lag time-series	Daily mean temperature over lag 0-14 in a quasi-Poisson DLNM, plus lag0-1 PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ , CO and O ₃ in single- and two-pollutant models	China
Reproductive & developmental					
Mbishi et al.	Early-life exposure to particulate matter and green spaces: Associations with infant growth	Environ Pollut	Prospective cohort	Perinatal PM ₁₀ and PM _{2.5} from the RIO detrended-kriging interpolation model; municipal vegetation split into low, high and total green space	Belgium

First author	Focus	Journal	Design	Metric	Region
Wei Q et al.	Prenatal exposure to PM _{2.5} components and fetal growth: identifying critical gestational weeks and key toxic components in a Chinese birth cohort	<i>Int J Epidemiol</i>	Prospective birth cohort	Weekly black carbon, organic matter, nitrate, sulfate and ammonium from the TAP real-time exposure model, entered into distributed lag models with GEE plus ML mixture analysis	China
Respiratory & allergic					
Wangupadcha et al.	Short-term effects of ambient PM _{2.5} on respiratory healthcare utilisation in older adults: a time-series study in an industrialising area of Central Thailand	<i>J Glob Health</i>	Time-series	Daily PM _{2.5} from the nearest government monitor, single-lag and moving-average negative binomial models per 10 µg/m ³	Thailand
Mechanistic toxicology					
Heo et al.	PM ₁₀ exposure impairs differentiation and induces inflammation in human small airway epithelial cells cultured at the air-liquid interface	<i>Tuberc Respir Dis (Seoul)</i>	In vitro exposure	Repeated PM ₁₀ dosing of differentiated air-liquid-interface cultures; TEER, lineage markers, surfactant proteins, cytokines, Notch and NF-κB signalling	South Korea
Ren Y et al.	Unveiling particulate matter-lung interactions from static models to dynamic lung-on-chips	<i>Research (Wash D C)</i>	Method review	Compares 2D culture, 3D bioprinted constructs, organoids and lung-on-chip platforms for reproducing PM deposition and breath-mimicking mechanical strain	Global / multi-region
Occupational & indoor					
Derwein et al.	Combined effects of face mask blocking, filtration, and ventilation strategy on exhaled air transmission and infection risk	<i>Build Environ</i>	Metadata only (no abstract)	Not retrievable: the publisher deposited no abstract to Crossref, Europe PMC or Semantic Scholar at index time	Not stated (metadata only)
Ibrahim	From complaint-driven response to preventive management: Strengthening indoor air quality management in Malaysian buildings	<i>Qeios (preprint)</i>	Commentary / methods letter	Argues for routine indoor monitoring of particulate matter and outdoor-to-indoor transport alongside CO ₂ , rather than complaint-triggered investigation under ICOP IAQ 2010	Malaysia
Kusi et al.	Mechanistic pathways of biochemical and toxicological responses to waste tyre combustion emissions: A systematic review	<i>Research Square (preprint)</i>	Systematic review (PRISMA)	Particulate matter, PAHs, VOCs and heavy metals from open waste-tyre burning, mapped to organ-specific biochemical and toxicological endpoints	Global / multi-region

First author	Focus	Journal	Design	Metric	Region
Wei J et al.	Welding fume exposure and respiratory innate immune activation: metal-rich particles, inflammatory biomarkers, and immune-informed occupational prevention	<i>Front Immunol</i>	Mechanistic review	Metal-rich fine and ultrafine welding-fume particles as a stimulus to airway epithelium, alveolar macrophages, neutrophils and circulating immune cells	Global / multi-region
Sensing, forecasting & instrumentation					
Cai J et al.	Scale-dependent drivers shape airborne microbiomes across the Shanghai subway network	<i>Environ Sci Technol</i>	Environmental surveillance campaign	Paired TSP ($\leq 100 \mu\text{m}$) and $\text{PM}_{2.5}$ sampling with microbial taxonomy, transcriptional potential, $\text{PM}_{2.5}$ chemical composition, microclimate and network topology	China
Ichikawa et al.	Tower- and drone-based BVOC observations in a suburban Tokyo forest: methodological insights and MEGAN comparison	<i>Atmos Chem Phys</i>	Multi-instrument field evaluation	Isoprene and monoterpene mixing ratios and emission fluxes at multiple heights on a 30 m flux tower, cross-checked by drone sampling and against MEGAN	Japan
Kim C-h et al.	OAV-based source characterization of designated odor compounds in long-term continuous monitoring data with severe missingness	<i>Atmosphere</i>	Monitoring-record analysis	22 designated odour compounds at 20 min resolution; block-holdout imputation with BiLSTM, BiGRU, TCN, Transformer and HistGradient-Boosting, then odour activity value and odour contribution	South Korea
Rosas et al.	Biomass burning and African dust influence on seasonal aerosol chemistry over the Western Caribbean (Part I): Atmospheric variability and human health	<i>Environ Sci Pollut Res Int</i>	Field campaign / remote sensing	Continuous $\text{PM}_{2.5}$, PM_{10} and particle-bound PAH with the $\text{PM}_{2.5} / \text{PM}_{10}$ ratio as a regime discriminator, joined to FIRMS fire detections and HYSPLIT back-trajectories	Mexico
Seong & Hoque	Influence of boundary layer conditions and surface properties on bacterial interaction with indoor built surfaces	<i>Front Pediatr</i>	Laboratory metrology	Attachment fraction and centrifugal detachment of <i>Corynebacterium</i> sp. on carpet, wood, metal and glass, with near-surface velocity fields by particle image velocimetry	Laboratory

First author	Focus	Journal	Design	Metric	Region
Zeng et al.	Decoupling between PM _{2.5} oxidative potential and light extinction: Roles of chemical composition and aerosol thermodynamic state	<i>J Hazard Mater</i>	Source apportionment	Volume-normalised oxidative potential and light extinction placed on a common quadrant framework, with water-soluble organic carbon, aerosol liquid water content, aerosol pH and PMF source factors as explanatory terms	Taiwan
Exposure assessment & modelling					
Arunachalam et al.	Residential gas combustion drives building-sector air pollution mortality in the U.S.	<i>Research Square (preprint)</i>	Health impact model	CMAQ Decoupled Direct Method sensitivities of NO ₂ , O ₃ and PM _{2.5} to building-sector precursor emissions, propagated through BenMAPR to premature deaths	United States
Leung et al.	Investigating the impacts of abandoned oil and gas wells on indoor air quality and health symptoms in Ontario, Canada: A pilot study	<i>Indoor Air</i>	Observational exposure study	13 passively sampled indoor VOCs (gas phase, not particulate) against inverse-distance-squared-weighted abandoned-well exposure within 3 km, with self-reported acute symptoms	Canada
Ullah et al.	Transboundary biomass-burning influence on springtime PM _{2.5} population exposure across southern China during 2015-2019	<i>Environ Pollut</i>	CTM simulation + source attribution	Population-weighted share of days above the WHO 2021 24 h guideline (15 µg/m ³) and China GB 3095-2012 Grade II (75 µg/m ³), with the Indo-China Peninsula fire contribution isolated by a paired GEOS-Chem run	China
Zhao C et al.	Incomplete reduction of particle-bound PAH toxicity under lower PM _{2.5} conditions: Evidence from Kunming, China	<i>Environ Pollut</i>	Source apportionment	16 priority PM _{2.5} -bound PAHs, BaP-equivalent burden, particle-normalised toxicity intensity (BaPeq/PM _{2.5}) and incremental lifetime cancer risk, split by lower- and higher-PM _{2.5} stage	China
Other clinical endpoints					
Ding E et al.	Climate factors, air pollutants, and extreme events shape influenza risk in China: a hospital-based case-crossover and machine-learning study	<i>Environ Sci Technol</i>	Time-series / case-crossover	Exposure-lag-response for meteorology, air pollution and compound extreme events by DLNM, variable ranking by ML, and an LSTM forecasting proof of concept for Beijing	China

First author	Focus	Journal	Design	Metric	Region
Wang C et al.	Age-dependent heterogeneity trends in the associations between urban green space and cardiovascular health in a humid-subtropical megacity	<i>Environ Sci Technol</i>	Cross-sectional + prediction model	Two-stage gradient-boosting plus structural equation model; air-quality pathways (NO ₂ , SO ₂ , PM _{2.5}) carry >75% of the relative contribution to model variance	China
Zheng M et al.	Spatiotemporal co-occurrence patterns and related risk factors of esophageal and lip and oral cavity cancers among older adults across 203 countries and territories	<i>Ann Med</i>	GBD secondary analysis	Occupational exposure to particulate matter, gases and fumes as one of 25 country-level exposures in negative binomial models with population offsets	Global / multi-region

Provenance and method

Entry window **4 September 2026**, single day, opened against a `last_entry_date` of 3 September. Sources queried on 5 September 2026 at 11:4x CDT; under this pipeline's 22:00-local rule, 4 September is the newest buildable date and 5 September is not opened. Exactly one day was owed and no gap remains. **The W36 weekly rollup (30 August–5 September) is therefore not built here:** it needs 5 September, which only the 23:00 run can open. **PubMed** E-utilities on [EDAT] 2026/09/04: **14** hits on the health axis, **3** on the instrumentation axis; the PubMed connector, run independently with looser field tags, returned 15 and 4 and surfaced no record the harvester had missed. **Europe PMC** `CREATION_DATE`: 24. **Crossref** by-ISSN, windowed on `created`: **57** deposits. **arXiv** answered without an HTTP 429 for the second consecutive run (149 kB of Atom) but its newest submission remains 2026-08-20, so the undated relevance query again contributed nothing; **OpenAlex** returned HTTP 429, the plan-gated date filter having been broken since 28 July. **Consensus** returned three low-cost-sensor calibration papers and all three were checked against Crossref `created` rather than assumed: two are out of window (2026-03-26, 2026-04-21) and the third is already in this corpus from an earlier issue — a sixth consecutive nil, now verified rather than inspected. After collapsing cross-source duplicates by PMID, DOI and normalised title, **91 unique records** were considered; **9 were already shipped** in earlier issues, **27 were screened in scope** and **55 were screened out**. Notable exclusions: two *International Journal of Population Data Science* conference abstracts, on London Underground commuting and on ambient air pollution and dementia — both squarely on topic, both with no DOI and no retrievable text; two *medRxiv* preprints on thermal variability and childhood respiratory mortality, where PM_{2.5} enters only as an adjustment covariate; an Apple Hearing Study analysis of noise and heart rate variability, methodologically close to distributed personal sensing but carrying no air-pollution metric; and a critical review of black-carbon marine carbon sequestration, which is atmospheric in origin and oceanographic in question. **Trial watch:** one new interventional registration, **NCT07805824** (cluster RCT, workplace heat and air-pollution mitigation, 20 restaurants and 200 oven workers, Najaf, Iraq; primary endpoint an objective gingival melanin index, sole secondary endpoint change in personal PM_{2.5} by gravimetric and real-time monitoring). `analyze_endpoints` was run because this is a new registration and not an update. Its phase label, start and completion dates are internally inconsistent and are recorded as unreliable. Logged to `state/trials.json` for the W36 weekly. The DOI gate passed all 27 records with **0 failures and 0 warnings**. All 24 distinct design labels were already present in the figure map, so no `plots.py` edit was made this run. Subtopic, design, particle-metric and geography labels are single-label by dominant contribution and are a judgement, not a controlled vocabulary. All DOI links resolve to the publisher of record; abstracts remain the copyright of their respective publishers.